HOT SPOTS AND RIGIDITY ON DIAGONAL FOLIATED DOMAINS IN $\mathbb{R}^n$

ABSTRACT. We prove the strict hot-spots property for an explicit family of smooth convex domains in every dimension $n \geq 2$. The domains are defined in a principal coordinate box by $\sum_{i=1}^{n-1} c_i \sin^2(y_i/\sqrt{2}) < \cos s$. For every $n \geq 3$, the dimension-independent condition $c_i \geq 2$ ensures that the first positive Neumann eigenvalue is simple and equals $(n+1)/2$, with eigenfunction $\sin s \prod_i \cos(y_i/\sqrt{2})$. A slightly wider coefficient range is proved in dimension two. We verify real-analytic boundary regularity, global convexity, and strictly positive principal curvatures, including the endpoint parameters and the two poles. The spectral identification combines an injective mean-gradient map with explicit comparison functions positive on transverse half-domains. The gradient leaves give a foliation away from the poles and allow independent changes of all diagonal parameters. The diameter is exactly π, while the diameter–volume ratio is unbounded. We also classify every connected C^2 domain in the principal box on whose boundary this prescribed function satisfies the Neumann condition. Such a domain is determined by a positive definite matrix, equal to the second fundamental form at either pole; coordinate reflections force the matrix to be diagonal. The proof reduces tangency to an exact homogeneity equation at the pole. We derive uniqueness from pole curvature data and characterize exact deformations preserving the eigenfunction. In dimensions $n \geq 3$, a convex, reflection-symmetric $C^{1,1}$ example shows that twice differentiable pole regularity is essential for the classification.

Draft status. This candidate proof was obtained by our team on October 18, 2026, through extensive dialogue-based training with AI. The team is currently working hard to verify all the proofs.

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1. Introduction and main results

Let $\Omega \subset \mathbb{R}^n$ be bounded, connected, and smooth. We number the Neumann eigenvalues, with multiplicities, as

$$0 = \mu_0(\Omega) < \mu_1(\Omega) \le \mu_2(\Omega) \le \cdots,$$

and write $E_1(\Omega)$ for the eigenspace of the first positive eigenvalue μ_1. All eigenfunctions are real valued. The strict hot-spots property considered here is

$$\min_{\partial\Omega} u < u(x) < \max_{\partial\Omega} u \qquad (x \in \Omega, \quad 0 \ne u \in E_1(\Omega)). \tag{1.1}$$

Our theorem gives the stronger conclusion that every such u has no interior critical point and has exactly two boundary critical points.

1.1. **Research context and the role of the construction.** For convex planar domains with two orthogonal axes of symmetry, the hot-spots property was proved by Jerison and Nadirashvili [3]. Klartag's analysis of Neumann eigenfunctions on convex bodies includes the nonvanishing of the mean gradient and its consequences for multiplicity and symmetry [5, Section 4]. Kennedy and Rohleder proved strong hot-spots results on higher-dimensional lip domains and related symmetric domains through a vector variational method [4]. Their higher-dimensional symmetry results have additional geometric hypotheses; coordinate-reflection symmetry by itself is not the theorem's hypothesis. The relation between gradient trajectories and Neumann boundary conditions is also studied in two-dimensional Neumann-domain theory, where identifying the spectral position is a separate problem [1].

Convexity alone does not imply the hot-spots property in arbitrary dimension: de Dios Pont constructed smooth convex counterexamples in all sufficiently large dimensions [2]. Thus a construction with a Neumann eigenfunction must still identify its spectral index. This is the essential spectral step in the present paper. All geometric and spectral arguments needed for the explicit family are supplied below; no general hot-spots or monotonicity theorem is used as an input.

The phrase *diagonal foliated domain* in this paper refers to the precise trigonometric family defined next. It is not an assumption that an arbitrary convex domain possesses a foliation with the same analytic properties.

1.2. **Definition and the hot-spots theorem.** Fix $n \ge 2$ and put $m = n-1$. Write $x = (s, y) \in \mathbb{R} \times \mathbb{R}^m$, and let

$$\mathcal{B} = \left(-\frac{\pi}{2}, \frac{\pi}{2}\right) \times \left(-\frac{\pi}{\sqrt{2}}, \frac{\pi}{\sqrt{2}}\right)^m.$$

For a vector $c = (c_1, \dots, c_m)$ with $c_i > 1$, define

$$F_c(s, y) = \sum_{i=1}^{m} c_i \sin^2\left(\frac{y_i}{\sqrt{2}}\right) - \cos s, \qquad \Omega_c = \{(s, y) \in \mathcal{B} : F_c(s, y) < 0\}. \tag{1.2}$$

The coordinate-box restriction is part of the definition; the full periodic sublevel set in $\mathbb{R}^n$ is not intended. Set

$$c_n^\star = \csc^2\left(\frac{\pi}{2\sqrt{n+1}}\right), \qquad \tau_n = \min\{2, c_n^\star\} = \begin{cases} \csc^2\left(\dfrac{\pi}{2\sqrt{3}}\right), & n = 2, \\ 2, & n \ge 3. \end{cases} \tag{1.3}$$

In particular $\tau_2 = 1.612099107496\ldots$, and $c_3^\star = 2$. The exact expressions, rather than the decimals, specify all endpoints.

Theorem 1.1 (Hot spots on diagonal foliated domains)**.** *Let $n \geq 2$ and suppose*

$$c_i \geq \tau_n \qquad (1 \leq i \leq n-1). \tag{1.4}$$

Then Ω_c is bounded and convex, with real-analytic boundary and strictly positive principal curvatures everywhere. Its first positive Neumann eigenvalue is simple, and

$$\mu_1(\Omega_c) = \Lambda_n := \frac{n+1}{2}, \qquad E_1(\Omega_c) = \operatorname{span}\{U\}, \qquad U(s,y) = \sin s \prod_{i=1}^{n-1} \cos\!\left(\frac{y_i}{\sqrt{2}}\right). \tag{1.5}$$

Moreover,

$$U_s > 0 \quad \text{on } \overline{\Omega}_c \setminus \{p_+, p_-\}, \qquad p_\pm = (\pm\pi/2, 0). \tag{1.6}$$

The points p_+ and p_- are the only critical points of U on $\overline{\Omega}_c$, and are its unique maximum and minimum, with values 1 and -1. They are nondegenerate extrema of its boundary restriction. Consequently every nonzero first eigenfunction satisfies (1.1).

The Euclidean diameter of Ω_c is exactly π. The ratio $d(\Omega_c)^n/|\Omega_c|$ is unbounded within the parameter range (1.4).

The conditions are independent in all $n-1$ diagonal parameters. In particular $c_i \geq 2$ suffices in both dimensions three and four, and in every higher dimension. The alternative sufficient condition $c_i \geq c_n^\star$, obtained from interval estimates below, is strictly more restrictive for $n \geq 4$. For example $c_4^\star = 2.395422833039\ldots$.

1.3. **Classification and rigidity.** The same explicit function also determines the possible smooth domains on which it satisfies the Neumann condition. The following geometric classification uses the fixed coordinate box $\mathcal{B}$ and the fixed ambient function U from (1.5).

Theorem 1.2 (Determination by the curvature at a pole)**.** *Let $\Omega \subset \mathcal{B}$ be a nonempty connected domain with C^2 boundary, and suppose $\partial_\nu U = 0$ on $\partial\Omega$. Then there is a unique symmetric positive definite matrix A with $(A^{-1})_{ii} < 1$ for every i such that*

$$\Omega = \left\{(s,y) \in \mathcal{B} : Y(y)^T A Y(y) < \cos s\right\}, \qquad Y_i(y) = \sin(y_i/\sqrt{2}). \tag{1.7}$$

Its boundary is real analytic, and both poles $p_\pm = (\pm\pi/2, 0)$ belong to the boundary. In the transverse coordinate basis, their outward second fundamental forms satisfy

$$\mathrm{II}_{p_+} = \mathrm{II}_{p_-} = A. \tag{1.8}$$

Thus the second fundamental form at one pole determines the whole domain. If Ω is invariant under every transverse coordinate reflection, then $A = \operatorname{diag}(c_1, \ldots, c_m)$ with $c_i > 1$, and the domain is exactly Ω_c.

The expanded classification, including its converse, is proved in Section 6. Under the coefficient bounds of Theorem 1.1, it applies to the simple first eigenfunction and yields rigidity within the hot-spots family. The classification itself imposes no convexity assumption and does not identify the spectral index for an arbitrary matrix A. The first eigenvalue and diameter alone leave shape freedom: all domains $\Omega_{a\mathbf{1}}$ with $a \geq 2$ have $\mu_1 = (n+1)/2$ and $d = \pi$, but their volumes vary strictly with a.

1.4. **Outline of the proof.** We first verify the geometry in the original Euclidean coordinates. This includes the positive tangential Hessian at $c_i = 2$, where the full defining Hessian can be singular, and an explicit smooth graph at each pole. Direct differentiation proves the Helmholtz equation and the Neumann condition for U. The quantities $\sin(y_i/\sqrt{2})/\sqrt{\cos s}$ are constant along its gradient curves, giving the promised foliation.

Next we prove mean-gradient injectivity and use coordinate parity to restrict the first eigenspace to coordinate-odd sectors. For a transverse coordinate, the comparison function

$$V_i(s, y) = \cos(s/2)\sin(\sqrt{2}y_i)\prod_{j\neq i}\cos(y_j/\sqrt{2})$$

is positive on $\Omega_c \cap \{y_i > 0\}$ and satisfies $-\Delta V_i = (\Lambda_n + 3/4)V_i$ and $\partial_\nu V_i = a_i V_i$ on the boundary, with $a_i \geq 0$ when $c_i \geq 2$. Green's identity against the odd ground state chosen positive on this half-domain gives the transverse lower bound $\lambda_i \geq \Lambda_n + 3/4$ in that coefficient range. Thus, when all $c_i \geq 2$, every first eigenfunction must be axially odd. Positivity of U on $s > 0$ identifies the axial ground state and proves simplicity. A separate one-dimensional comparison supplies the planar extension in (1.3).

After calculating the diameter, volume, and independent parameter changes, we prove the classification theorem. The Neumann condition first excludes contact with the transverse faces of the principal box and makes the domain a union of complete gradient leaves. Near the upper pole, write the boundary as $q = g(Y)$, where $q = \cos s$. Tangency gives the Euler equation $Y \cdot \nabla g = 2g$. Twice differentiability at the origin then forces the exact identity $g(Y) = Y^T AY$. Transport along the leaves extends this local identity to the whole domain. We derive the curvature, congruence, and deformation consequences, and finish with the local version for arbitrary label sets and, for $n \geq 3$, a convex $C^{1,1}$ example showing the necessity of the pole regularity assumption.

2. Geometry and the explicit gradient foliation

Throughout this section the coefficients satisfy $c_i \geq \tau_n$ as in Theorem 1.1. Write

$$Y_i = \sin(y_i/\sqrt{2}), \qquad C_i = \cos(y_i/\sqrt{2}), \qquad P(y) = \prod_{i=1}^{m} C_i. \tag{2.1}$$

The functions C_i are strictly positive on the principal box.

Proposition 2.1 (Regularity, convexity, and the endpoint geometry)**.** *The set Ω_c is a bounded connected convex domain with real-analytic boundary and strictly positive principal curvatures. At either pole its principal curvatures, with the outward normal convention, are $c_1, \ldots, c_m$.*

Proof. We first establish connectedness and smoothness for all $c_i > 1$. In the coordinates (s, Y), consider

$$K_c = \left\{(s, Y) : |s| \leq \frac{\pi}{2},\ \sum_i c_i Y_i^2 \leq \cos s\right\}.$$

The function $\widetilde{F}_c(s, Y) = \sum_i c_i Y_i^2 - \cos s$ is convex on the strip $|s| \leq \pi/2$, and $\widetilde{F}_c(0, 0) = -1$. Thus K_c is compact and convex with nonempty interior. Its interior consists exactly of the points with $|s| < \pi/2$ and $\widetilde{F}_c < 0$: at a zero of $\widetilde{F}_c$ with $|s| < \pi/2$, one has $Y \neq 0$, and a small increase of Y along its ray leaves K_c. Moreover, every point of K_c is approached from its interior by taking convex combinations with $(0, 0)$, since convexity makes $\widetilde{F}_c$ strictly negative on those combinations. The analytic change of variables $y_i = \sqrt{2}\arcsin Y_i$ is a diffeomorphism on a neighborhood of K_c, because $|Y_i| \leq c_i^{-1/2} < 1$. It maps $\operatorname{int} K_c$ onto Ω_c and K_c onto $\overline{\Omega}_c$. In particular Ω_c is bounded, nonempty, and connected. Its closure is disjoint from the transverse faces of the principal box and meets the axial faces only at p_+ and p_-.

In the original coordinates,

$$(F_c)_s = \sin s, \qquad (F_c)_{y_i} = \sqrt{2}c_i Y_i C_i, \tag{2.2}$$

and

$$D^2 F_c = \operatorname{diag}(\cos s, c_1(1-2Y_1^2), \dots, c_m(1-2Y_m^2)). \tag{2.3}$$

The gradient is nonzero on the whole boundary. At $s \neq 0$ the axial component is nonzero. At $s = 0$ the boundary equation $\sum_i c_i Y_i^2 = 1$ supplies a nonzero transverse component. The analytic implicit function theorem therefore supplies a real-analytic hypersurface at every boundary point. Near a pole there is no additional boundary contributed by the edge of the principal box: for $|s| > \pi/2$ sufficiently close to that pole, $F_c > 0$. The local boundary is exactly the level $F_c = 0$.

Suppose first that all $c_i \geq 2$, as is required in dimensions $n \geq 3$. Set $b_i = \sqrt{2}\arcsin(c_i^{-1/2})$. On the convex rectangle

$$Q_c = [-\pi/2, \pi/2] \times \prod_i [-b_i, b_i],$$

one has $Y_i^2 \leq 1/c_i$ and hence $c_i(1-2Y_i^2) \geq c_i - 2 \geq 0$. Thus F_c is convex on Q_c. Since $\Omega_c = \{F_c < 0\} \cap \operatorname{int} Q_c$, it is globally convex in the Euclidean coordinates.

For a regular level hypersurface bounding a sublevel set, the outward second fundamental form is

$$\mathrm{II}(T,T) = \frac{D^2 F_c[T,T]}{|\nabla F_c|}, \qquad T \cdot \nabla F_c = 0. \tag{2.4}$$

At a nonpolar boundary point $\cos s > 0$. The only possible zero transverse diagonal entry in (2.3) occurs if $c_i = 2$ and $Y_i^2 = 1/2$. The boundary identity then forces $s = 0$ and $Y_j = 0$ for every $j \neq i$. There is only one zero Hessian direction, namely e_{y_i}, and (2.2) shows that this is the normal direction. The Hessian is therefore positive definite on the tangent space at every nonpolar boundary point, including the endpoint cases $c_i = 2$. The pole curvatures will be computed below.

It remains to cover the planar coefficients $1 < c < 2$. Here $m = 1$. On the boundary put $q = \cos s = cY^2$ and take the nonzero tangent vector $T = ((F_c)_y, -(F_c)_s)$. Using $Y^2 = q/c$, $C^2 = 1 - q/c$, and $\sin^2 s = 1 - q^2$ in (2.2)–(2.3) gives

$$\begin{aligned} D^2 F_c[T,T] &= 2cq^2 - 2q^3 + (c-2q)(1-q^2) \\ &= c(1+q^2) - 2q \\ &= (c-1)(1+q^2) + (q-1)^2 > 0. \end{aligned} \tag{2.5}$$

This proves positive curvature at every nonpolar point in the remaining planar range. To verify global convexity without a topological convexity theorem, write $b = \sqrt{2}\arcsin(c^{-1/2})$ and

$$a(y) = \arccos\!\big(c\sin^2(y/\sqrt{2})\big), \qquad |y| < b.$$

Then $\Omega_c = \{(s,y) : |y| < b,\ -a(y) < s < a(y)\}$. Implicit differentiation of $F_c(a(y), y) = 0$ yields

$$a'(y) = -\frac{(F_c)_y}{(F_c)_s}, \qquad a''(y) = -\frac{c(1+q^2) - 2q}{\sin^3 a(y)} < 0 \qquad (|y| < b).$$

Here the derivatives of F_c are evaluated at $(a(y), y)$, $q = \cos a(y)$, and $0 < a(y) \leq \pi/2$, so the denominator is strictly positive. The second identity follows by differentiating the first and using $(a', 1) = -T/(F_c)_s$ in the preceding tangential Hessian formula. The function extends continuously with $a(\pm b) = 0$. It is concave on its closed interval, so its hypograph and the epigraph of $-a$ are convex. Their intersection is $\overline{\Omega}_c$, whose interior is Ω_c. This proves global convexity in the remaining planar range. Smoothness at $|y| = b$ follows from the defining function, not from assuming a finite derivative for this graph.

Finally, near the upper pole put $t = \pi/2 - s$. The boundary equation is

$$\sin t = \sum_i c_i \sin^2(y_i/\sqrt{2}), \qquad t = \arcsin\left(\sum_i c_i \sin^2(y_i/\sqrt{2})\right).$$

It is a single analytic graph near $y = 0$, and its expansion is

$$t = \frac{1}{2}\sum_i c_i y_i^2 - \frac{1}{12}\sum_i c_i y_i^4 + O(|y|^6). \tag{2.6}$$

At the pole the normal is axial and the graph has zero gradient; its positive Hessian is $\mathrm{diag}(c_i)$. The lower pole has the same expansion in $t = \pi/2 + s$. This also verifies the curvature values claimed in the proposition. □

Proposition 2.2 (An exact Neumann eigenpair)**.** *For the domain in Theorem 1.1, the function U in (1.5) satisfies*

$$-\Delta U = \Lambda_n U \quad \text{in } \Omega_c, \qquad \partial_\nu U = 0 \quad \text{on } \partial\Omega_c, \qquad \int_{\Omega_c} U = 0.$$

It is positive on $s > 0$, and has the derivative and extremum properties in (1.6). At this point its spectral index has not yet been identified.

Proof. The axial factor has frequency 1, and each transverse factor has squared frequency $1/2$. Hence $-\Delta U = (1 + m/2)U = \Lambda_n U$. Furthermore,

$$U_s = \cos s\, P, \qquad U_{y_i} = -\frac{\sin s}{\sqrt{2}}\, P \frac{Y_i}{C_i}.$$

These identities and (2.2) give the exact identity

$$\nabla F_c \cdot \nabla U = \sin s\, P\left(\cos s - \sum_i c_i Y_i^2\right) = -\sin s\, P F_c. \tag{2.7}$$

Thus the normal derivative is zero on the full boundary, including the poles, where all expressions are smooth. Reflection in $s = 0$ preserves the domain and changes the sign of U, proving its zero mean. It follows already that $\mu_1(\Omega_c) \le \Lambda_n$.

Every nonpolar point of the closure has $|s| < \pi/2$ and all $C_i > 0$; therefore $U_s = \cos s\, P > 0$ there. At $p_\pm$ the gradient is zero and the values are ± 1. Since $|U| \le |\sin s| \le 1$ on the closure, these extrema are unique. The Neumann condition makes ∇U tangent to the boundary, so it is also the gradient of the boundary restriction. Thus that restriction has no critical points other than the two poles. Finally

$$D^2U(p_+) = \mathrm{diag}(-1, -1/2, \dots, -1/2), \qquad D^2U(p_-) = -D^2U(p_+).$$

The boundary tangent spaces are $ds = 0$. Because $\nabla U = 0$ at the poles, the Hessian of the boundary restriction is exactly the restriction of this ambient Hessian. It is definite, proving nondegeneracy of the two boundary extrema. □

Proposition 2.3 (The gradient leaves)**.** *Define the ellipsoidal label set*

$$D_c = \left\{\eta \in \mathbb{R}^m : \sum_i c_i \eta_i^2 < 1\right\}.$$

The map

$$\Phi_c(s, \eta) = \left(s, \sqrt{2}\arcsin(\eta_1\sqrt{\cos s}), \dots, \sqrt{2}\arcsin(\eta_m\sqrt{\cos s})\right) \tag{2.8}$$

is a smooth diffeomorphism from $(-\pi/2, \pi/2) \times D_c$ onto Ω_c. For a fixed η, its image is a gradient trajectory of U up to an increasing reparametrization. Labels in ∂D_c parametrize the nonpolar boundary leaves. Each leaf has both poles in its geometric closure.

Proof. The inverse map is explicit:

$$(s, y) \longmapsto \left(s, \frac{Y_1}{\sqrt{\cos s}}, \dots, \frac{Y_m}{\sqrt{\cos s}} \right).$$

For $\eta \in \overline{D}_c$ one has $|\eta_i| \le c_i^{-1/2} < 1$, so all inverse sine arguments stay strictly between -1 and 1. Substitution gives

$$F_c(\Phi_c(s, \eta)) = \cos s \left(\sum_i c_i \eta_i^2 - 1 \right).$$

Thus the interior and boundary inequalities correspond exactly. Both the displayed map and its inverse are smooth for $|s| < \pi/2$. The same formulas extend smoothly to a neighborhood of every point with $\eta \in \partial D_c$ and $|s| < \pi/2$. This proves the diffeomorphism and the parametrization of the nonpolar boundary.

Fix $\eta \in \overline{D}_c$ and write $z(s) = \Phi_c(s, \eta) = (s, y(s))$. Differentiation gives

$$\frac{dY_i}{ds} = -\frac{1}{2} \tan s \, Y_i, \qquad \frac{dy_i}{ds} = -\frac{\tan s}{\sqrt{2}} \frac{Y_i}{C_i} = \frac{U_{y_i}}{U_s}.$$

Consequently $z'(s) = \nabla U(z(s))/U_s(z(s))$. Since $U_s = \cos s \, P > 0$, define the strictly increasing flow time

$$\tau(s) = \int_0^s \frac{\mathrm{d}\sigma}{\cos \sigma \, P(y(\sigma))}.$$

The chain rule gives $dz/d\tau = \nabla U(z)$. Conversely, along any gradient trajectory in $\overline{\Omega}_c \setminus \{p_+, p_-\}$, s is strictly increasing, and the same differential equation for Y_i makes $Y_i/\sqrt{\cos s}$ constant. Uniqueness for this smooth vector field therefore identifies its maximal trajectory with the corresponding fixed-label curve.

As $s \to \pm\pi/2$, $y(s) \to 0$ and $P(y(s)) \to 1$. Moreover, $0 < P \le 1$, so comparison with $\int \mathrm{d}s/\cos s$ shows that $\tau(s) \to \pm\infty$ at the respective endpoints. Thus these are complete gradient trajectories, with the two poles as their respective limiting points. The leaf coordinates remain smooth away from the poles and do not extend there as nonsingular coordinates. □

3. Mean gradients and coordinate symmetry

This section applies to any bounded connected convex domain $\Omega \subset \mathbb{R}^n$ with C^∞ boundary and positive principal curvatures. These properties have already been proved for the domains under consideration. We use the outward second fundamental form $\mathrm{II}(S, T) = \langle D_S \nu, T \rangle$, so that a sphere has positive curvature. Put $\mu = \mu_1(\Omega) > 0$.

By the variational characterization of μ, every $f \in H^1(\Omega)$ satisfies the Neumann Poincaré inequality

$$\int_\Omega |\nabla f|^2 \ge \mu \int_\Omega |f - f_\Omega|^2, \qquad f_\Omega = \frac{1}{|\Omega|} \int_\Omega f. \tag{3.1}$$

For a mean-zero function, equality holds precisely when it belongs to $E_1(\Omega)$, with the zero function permitted. Indeed, if equality holds, apply (3.1) to $f + t(\psi - \psi_\Omega)$ for arbitrary $\psi \in H^1(\Omega)$. The difference between the two sides is nonnegative for every real t and vanishes at $t = 0$, so its derivative there is zero. Since $\int_\Omega f = 0$, this gives

$$\int_\Omega \nabla f \cdot \nabla \psi = \mu \int_\Omega f \psi \qquad (\psi \in H^1(\Omega)).$$

This is the full weak Neumann eigenvalue equation. Conversely, testing that equation with $\psi = f$ proves equality in (3.1).

Lemma 3.1 (Injectivity of the mean gradient)**.** *The map*

$$J : E_1(\Omega) \longrightarrow \mathbb{R}^n, \qquad J(u) = \int_\Omega \nabla u$$

is injective. In particular $\dim E_1(\Omega) \le n$.

Proof. Smooth Neumann regularity gives $u \in C^\infty(\overline{\Omega})$. An integration identity for its Hessian is

$$\int_\Omega |D^2 u|^2 + \int_{\partial\Omega} \mathrm{II}(\nabla_\tau u, \nabla_\tau u) = \int_\Omega (\Delta u)^2 = \mu \int_\Omega |\nabla u|^2. \tag{3.2}$$

To verify the boundary sign, integrate the divergence identity

$$(\Delta u)^2 - |D^2 u|^2 = \mathrm{div}\big((\Delta u)\nabla u - D^2 u \nabla u\big).$$

The boundary flux is $-D^2 u(\nabla_\tau u, \nu)$ because $\partial_\nu u = 0$. For every tangent vector T, differentiating the Neumann condition along the boundary gives

$$0 = T(\nabla u \cdot \nu) = D^2 u(T, \nu) + \langle \nabla_\tau u, D_T \nu \rangle.$$

Taking $T = \nabla_\tau u$ identifies the flux with $\mathrm{II}(\nabla_\tau u, \nabla_\tau u)$, with the stated outward-normal sign convention. The last equality in (3.2) uses $-\Delta u = \mu u$ and the Neumann energy identity.

For $w_i = \partial_i u$, let the Poincaré deficits be

$$\mathcal{D}_i = \int_\Omega |\nabla w_i|^2 - \mu \int_\Omega |w_i - (w_i)_\Omega|^2 \ge 0.$$

Since $\sum_i \int_\Omega |\nabla w_i|^2 = \int_\Omega |D^2 u|^2$ and $\sum_i |(w_i)_\Omega|^2 = |J(u)|^2/|\Omega|^2$, (3.2) gives the exact relation

$$\frac{\mu}{|\Omega|}|J(u)|^2 = \int_{\partial\Omega} \mathrm{II}(\nabla_\tau u, \nabla_\tau u) + \sum_{i=1}^n \mathcal{D}_i. \tag{3.3}$$

If $J(u) = 0$, all terms on the right are nonnegative, so the boundary integral and every $\mathcal{D}_i$ vanish. Positivity of II gives $\nabla_\tau u = 0$ everywhere on the boundary: the integrand is continuous, nonnegative, and positive wherever the tangential gradient is nonzero. The Neumann condition then gives $\nabla u = 0$ there. Also $(w_i)_\Omega = 0$, and $\mathcal{D}_i = 0$ is equality in (3.1). Its equality characterization therefore gives $w_i \in E_1(\Omega)$, with zero allowed. Thus $w_i = 0$ on the boundary by $\nabla u = 0$, while $\partial_\nu w_i = 0$ follows from its Neumann eigenvalue equation.

To conclude from these zero Cauchy data, extend w_i by zero to a function W_i on $\mathbb{R}^n$. For any $\psi \in C_c^\infty(\mathbb{R}^n)$, Green's second identity and the two zero boundary data give

$$\int_{\mathbb{R}^n} W_i(-\Delta\psi - \mu\psi) = \int_\Omega w_i(-\Delta\psi - \mu\psi) = 0.$$

Thus $(-\Delta - \mu)W_i = 0$ distributionally on $\mathbb{R}^n$. Interior elliptic regularity and analyticity for this constant-coefficient equation make W_i real analytic on $\mathbb{R}^n$. It vanishes on the nonempty exterior open set $\mathbb{R}^n \setminus \overline{\Omega}$, so it vanishes identically. Hence $\nabla u = 0$ in Ω; connectedness and the zero mean of u give $u = 0$. This proves injectivity. The dimension bound follows because J is a linear injection into $\mathbb{R}^n$. □

Corollary 3.2 (The only possible first-eigenfunction parities)**.** *Suppose in addition that* Ω *is invariant under reflection in each coordinate hyperplane. Its first eigenspace is a direct sum of subspaces whose functions are odd in exactly one coordinate and even in all the others. Every first eigenfunction is centrally odd.*

Proof. Let r_j be reflection in $x_j = 0$ and put $R_j f = f \circ r_j$. A change of variables under r_j preserves both the gradient and L^2 pairings in the weak Neumann equation. Thus each R_j preserves $E_1(\Omega)$. The operators R_j commute and satisfy $R_j^2 = I$. For $\varepsilon = (\varepsilon_1, \dots, \varepsilon_n) \in \{-1, 1\}^n$, define

$$f_\varepsilon = \prod_{j=1}^{n} \frac{I + \varepsilon_j R_j}{2} f.$$

Then $f = \sum_\varepsilon f_\varepsilon$ and $R_j f_\varepsilon = \varepsilon_j f_\varepsilon$. Each component belongs to $E_1(\Omega)$. Components with different parity vectors are L^2-orthogonal, because their product is odd in at least one coordinate; hence this is a direct-sum decomposition.

For a component f, the integral of $\partial_i f$ can be nonzero only if f is odd in coordinate i and even in every other coordinate: otherwise that derivative remains odd in at least one coordinate, and reflection makes its integral zero. All other parity patterns therefore have $J(f) = 0$ and vanish by Lemma 3.1. The surviving patterns are exactly the stated ones, and each is odd under simultaneous inversion of all coordinates. Their sums are centrally odd as well. □

4. Identification of the first positive Neumann eigenvalue

Let $m = n - 1$, let Ω_c be the domain of Proposition 2.1, and assume throughout this section that

$$c_i \geq \tau_n := \min\{2, c_n^\star\}, \qquad c_n^\star := \csc^2\left(\frac{\pi}{2\sqrt{n+1}}\right) \qquad (1 \leq i \leq m). \tag{4.1}$$

Thus $\tau_2 = c_2^\star < 2$, while $\tau_n = 2$ for $n \geq 3$. The domain is smooth, bounded, convex, and invariant under reflection in every coordinate hyperplane. Proposition 2.2 provides the Neumann eigenfunction

$$U(s, y) = \sin s \prod_{j=1}^{m} \cos\left(\frac{y_j}{\sqrt{2}}\right), \qquad -\Delta U = \Lambda_n U, \qquad \Lambda_n = \frac{n+1}{2}. \tag{4.2}$$

We prove that this is the simple first positive eigenpair. Two elementary estimates exclude the transverse odd sectors: a one-dimensional half-chord inequality, and an explicit trigonometric comparison function. The second gives the dimension-independent threshold 2, while the first gives the smaller threshold in dimension two.

For a coordinate $z \in \{s, y_1, \dots, y_m\}$, let

$$\mathcal{H}_z = \{f \in H^1(\Omega_c) : f \text{ is odd in } z\}, \qquad \lambda_z = \inf_{0 \neq f \in \mathcal{H}_z} \frac{\int_{\Omega_c} |\nabla f|^2}{\int_{\Omega_c} |f|^2}.$$

The subspace $\mathcal{H}_z$ is nonzero, since it contains the coordinate function z, and is closed, since reflection is a bounded linear operator on $H^1(\Omega_c)$. A minimizing sequence with L^2 norm one is bounded in H^1. Weak H^1 and strong L^2 compactness give a limit in $\mathcal{H}_z$ with L^2 norm one, and lower semicontinuity of the energy shows that it attains λ_z. Every odd function has zero mean, so (3.1) gives $\lambda_z \geq \mu_1(\Omega_c) > 0$. We write $\lambda_\parallel = \lambda_s$ and $\lambda_i = \lambda_{y_i}$.

4.1. An elementary inequality on a half-interval.

Lemma 4.1 (Dirichlet–Neumann inequality)**.** *If $h > 0$ and $g \in H^1(0, h)$ has trace $g(0) = 0$, then*

$$\int_0^h |g'(t)|^2 \, \mathrm{d}t \geq \frac{\pi^2}{4h^2} \int_0^h |g(t)|^2 \, \mathrm{d}t. \tag{4.3}$$

No boundary condition on $g(h)$ or $g'(h)$ is imposed on the test function. Equality holds precisely for scalar multiples of $\sin(\pi t/(2h))$.

Proof. Minimize the quotient on the closed subspace $\{g \in H^1(0,h) : g(0) = 0\}$ with L^2 norm one. Compactness gives a minimizer, and the elementary estimate $|g(t)| \le \sqrt{t}\,\|g'\|_{L^2(0,h)}$ shows that its eigenvalue is positive. The Euler–Lagrange equation and its natural boundary condition follow from

$$\int_0^h g'\psi' = \lambda \int_0^h g\psi \qquad (\psi \in H^1(0,h),\ \psi(0) = 0).$$

Compactly supported tests give $-g'' = \lambda g$ distributionally, and the one-dimensional equation makes g smooth up to the endpoints. Integration by parts then gives $g'(h)\psi(h) = 0$; the value $\psi(h)$ is arbitrary. Thus

$$-g'' = \lambda g, \qquad g(0) = 0, \qquad g'(h) = 0.$$

Writing $\lambda = k^2 > 0$, a nonzero solution is proportional to $\sin(kt)$ and satisfies $\cos(kh) = 0$. The smallest possible eigenvalue is therefore $\pi^2/(4h^2)$, and the corresponding sine function attains it. This proves the inequality and its equality case. □

4.2. **Strict exclusion of every transverse odd sector.** Fix $i \in \{1, \dots, m\}$. Denote the remaining coordinates by $\widehat{y}_i$, and let D_i be the projection of Ω_c onto the $(s, \widehat{y}_i)$ variables. For $(s, \widehat{y}_i) \in D_i$, put

$$A_i(s, \widehat{y}_i) = \cos s - \sum_{j \neq i} c_j \sin^2\left(\frac{y_j}{\sqrt{2}}\right), \tag{4.4}$$

$$h_i(s, \widehat{y}_i) = \sqrt{2}\arcsin\sqrt{\frac{A_i(s, \widehat{y}_i)}{c_i}}. \tag{4.5}$$

The transverse section of Ω_c over that point is exactly $-h_i < y_i < h_i$. Indeed, $0 < A_i \le 1$, and $\sin^2(t/\sqrt{2})$ is strictly increasing for $0 < t < \pi/\sqrt{2}$. Also $c_i > 1$, so the section endpoints occur strictly inside the principal coordinate box.

Set

$$H_i = \sqrt{2}\arcsin(c_i^{-1/2}), \qquad H_n^\star = \frac{\pi}{\sqrt{2(n+1)}}. \tag{4.6}$$

Whenever $c_i \ge c_n^\star$, the parameter condition gives

$$0 < h_i(s, \widehat{y}_i) \le H_i \le H_n^\star, \qquad \frac{\pi^2}{4(H_n^\star)^2} = \Lambda_n. \tag{4.7}$$

There is a useful strictness statement at the endpoint of the parameter range. Equality $h_i = H_n^\star$ can occur only when

$$c_i = c_n^\star, \qquad s = 0, \qquad y_j = 0 \quad (j \neq i). \tag{4.8}$$

In fact, equality requires $A_i = 1$; because $\cos s \le 1$ and all the subtracted terms in (4.4) are nonnegative, this forces exactly the stated coordinate values within the principal box. The exceptional base point in (4.8) has $(n-1)$-dimensional measure zero, also when $n = 2$.

Lemma 4.2 (Strict transverse odd-sector bound)**.** *If $c_i \ge c_n^\star$, every nonzero $f \in H^1(\Omega_c)$ which is odd in y_i satisfies*

$$\int_{\Omega_c} |\nabla f|^2 \, \mathrm{d}x > \Lambda_n \int_{\Omega_c} |f|^2 \, \mathrm{d}x. \tag{4.9}$$

In particular, $\lambda_i > \Lambda_n$.

Proof. Sobolev slicing gives an H^1 function on almost every interval $(-h_i, h_i)$. The section is odd and its one-dimensional continuous representative therefore vanishes at $y_i = 0$. Apply Lemma 4.1 on its positive half and use odd reflection on its negative half. Fubini's theorem yields

$$\begin{aligned}\int_{\Omega_c} |\nabla f|^2 \, \mathrm{d}x &\geq \int_{D_i} \int_{-h_i}^{h_i} |\partial_{y_i} f|^2 \, \mathrm{d}y_i \, \mathrm{d}(s, \widehat{y}_i) \\ &\geq \int_{D_i} \frac{\pi^2}{4h_i(s, \widehat{y}_i)^2} \int_{-h_i}^{h_i} |f|^2 \, \mathrm{d}y_i \, \mathrm{d}(s, \widehat{y}_i). \end{aligned} \tag{4.10}$$

By (4.7)– (4.8), the coefficient in the last integral is strictly greater than Λ_n for almost every base point. A nonzero f has nonzero L^2 mass on a positive-measure set of base points. Therefore the last integral is strictly greater than $\Lambda_n \|f\|^2_{L^2(\Omega_c)}$, which proves (4.9). The weighted integral is finite by (4.10), so this strictness argument involves no subtraction of infinite quantities.

The odd-sector minimum is attained by the compactness argument at the start of the section. Applying (4.9) to a minimizer proves $\lambda_i > \Lambda_n$; attainment is what allows passage from strictness for each test function to strictness of the infimum. □

4.3. **The axial odd ground state.**

Lemma 4.3 (Odd ground states and axial simplicity)**.** *The least eigenvalue in any one-coordinate-odd subspace is simple within that subspace. Its eigenfunction can be chosen strictly positive on the corresponding positive half-domain and is smooth on $\overline{\Omega_c}$. One has $\lambda_\parallel = \Lambda_n$. This eigenvalue is simple in the s-odd subspace, and its eigenfunctions are the scalar multiples of U in (4.2).*

Proof. Write $\Omega_+ = \Omega_c \cap \{s > 0\}$ and $\Gamma_0 = \Omega_c \cap \{s = 0\}$. An s-odd H^1 function has zero trace on Γ_0: its two traces agree because it is an H^1 function across that interior hyperplane, while reflection changes their signs. Conversely, an $H^1(\Omega_+)$ function with zero trace on Γ_0 has an odd H^1 extension to Ω_c. The matching zero traces eliminate an interface jump in the distributional first derivatives, so the extension belongs to $H^1(\Omega_c)$. Thus restriction identifies the odd variational problem with the mixed problem having zero Dirichlet trace on Γ_0 and natural Neumann data on the exterior boundary. Odd extension doubles both the Dirichlet energy and the squared L^2 norm; restriction halves them, so both operations preserve the Rayleigh quotient. The positive half-domain is nonempty and connected by convexity.

Take an s-odd minimizer, whose existence was established above. Replace its restriction to Ω_+ by its absolute value and extend oddly. The Sobolev chain rule and the zero trace on Γ_0 show that this preserves admissibility and both quadratic integrals, so the resulting function, denoted by ϕ, is also a minimizer. Its Euler–Lagrange equation with eigenvalue $\lambda_\parallel$ holds for every odd test function. For every even test function both its gradient pairing and its L^2 pairing with this odd minimizer vanish by reflection. Decomposing an arbitrary H^1 test into its odd and even parts proves the weak Neumann eigenvalue equation on the full domain. Full-domain Neumann regularity makes the minimizer smooth on $\overline{\Omega_c}$, including where the two portions of the mixed boundary meet. The strong minimum principle, applied to the nonnegative function on Ω_+ satisfying $\Delta\phi = -\lambda_\parallel \phi \leq 0$, makes it strictly positive there.

For any nonzero lowest odd eigenfunction w, the same construction shows that $|w|$ is strictly positive on Ω_+. Smoothness of w and connectedness of Ω_+ then force w to have one strict sign there. Two linearly independent lowest odd eigenfunctions would have a nonzero linear combination orthogonal to a positive one on Ω_+, contradicting this sign property. Thus the odd-sector ground state is simple. The same argument applies with any transverse coordinate y_i in place of s, proving the first assertion for every coordinate.

The explicit U is odd in s and positive on the positive half-domain: $\sin s > 0$ there and every cosine factor is positive inside the principal box. Its inner product with the positive odd ground

state is therefore positive on the full domain. Neumann eigenfunctions for distinct eigenvalues are orthogonal, as follows by testing their weak equations against one another. Consequently the eigenvalue of U equals $\lambda_{\|}$, and simplicity proves the assertion. □

4.4. **A dimension-independent transverse comparison.**

Lemma 4.4 (An explicit odd comparison function)**.** *If $c_i \geq 2$, then the least y_i-odd eigenvalue satisfies*

$$\lambda_i \geq \Lambda_n + \frac{3}{4}. \tag{4.11}$$

Proof. Set

$$V_i(s,y) = \cos\left(\frac{s}{2}\right)\sin(\sqrt{2}\,y_i)\prod_{j\neq i}\cos\left(\frac{y_j}{\sqrt{2}}\right). \tag{4.12}$$

The function is smooth and odd in y_i. In the half-domain $y_i > 0$ it is strictly positive. Indeed, the other cosine factors are positive in the principal box, and $c_i \geq 2$ gives

$$0 < \sqrt{2}\,y_i < 2\arcsin(c_i^{-1/2}) \leq \frac{\pi}{2}.$$

Direct differentiation in the separate coordinates yields

$$-\Delta V_i = \left(\frac{1}{4} + 2 + \frac{m-1}{2}\right)V_i = \left(\Lambda_n + \frac{3}{4}\right)V_i. \tag{4.13}$$

We next check its outward normal derivative, using the defining function F_c and coordinates from Section 2. Put

$$q = \cos s, \qquad q_j = c_j Y_j^2.$$

Product differentiation gives the identity

$$\begin{aligned}\nabla F_c \cdot \nabla V_i &= \left(-\frac{1}{2}(1-q) + c_i - 2q_i - \sum_{j\neq i} q_j\right)V_i \\ &= \left(c_i - \frac{1}{2} - \frac{q}{2} - q_i - F_c\right)V_i.\end{aligned}$$

Here $\sin s \sin(s/2) = (1-q)\cos(s/2)$ gives the axial term, and $\sin(\sqrt{2}y_i) = 2Y_iC_i$ gives the ith transverse term. Thus no division by V_i is used, and the identity holds on $\overline{\Omega}_c$, including every zero of V_i. On $\partial\Omega_c$, one has $F_c = 0$, $\sum_j q_j = q$, and $0 \leq q_i \leq q \leq 1$. By Proposition 2.1, $\nabla F_c \neq 0$ on the boundary and $\nu = \nabla F_c/|\nabla F_c|$. We obtain

$$\partial_\nu V_i = \frac{c_i - \frac{1}{2} - \frac{q}{2} - q_i}{|\nabla F_c|}\,V_i, \qquad c_i - \frac{1}{2} - \frac{q}{2} - q_i \geq c_i - 2 \geq 0. \tag{4.14}$$

Let ϕ_i be the smooth y_i-odd ground state from Lemma 4.3, chosen positive on $y_i > 0$. Then $V_i\phi_i > 0$ in each open half-domain, and their product is nonnegative on the boundary by continuity. Thus (4.14) gives $\phi_i\partial_\nu V_i \geq 0$ on the entire boundary, including the half where both odd functions change sign. Green's identity on the full smooth domain gives

$$\begin{aligned}\left(\Lambda_n + \frac{3}{4} - \lambda_i\right)\int_{\Omega_c} V_i\phi_i \,\mathrm{d}x &= \int_{\Omega_c}\left((-\Delta V_i)\phi_i - V_i(-\Delta\phi_i)\right)\,\mathrm{d}x \\ &= -\int_{\partial\Omega_c}\phi_i\partial_\nu V_i \,\mathrm{d}\sigma \leq 0.\end{aligned}$$

Here $\partial_\nu\phi_i = 0$, and the integral of $V_i\phi_i$ is strictly positive. This proves (4.11). The calculation uses the full smooth domain, so no assertion about corner regularity of a mixed problem is needed. □

Proposition 4.5 (The simple first positive eigenpair)**.** *Under* (4.1)*,*

$$\mu_1(\Omega_c) = \Lambda_n = \frac{n+1}{2}, \qquad E_1(\Omega_c) = \operatorname{span}\{U\}. \tag{4.15}$$

This includes equality in any or all of the parameter bounds $c_i \geq \tau_n$, and it holds for every $n \geq 2$.

Proof. Proposition 2.2 gives $\mu_1(\Omega_c) \leq \Lambda_n$. By Corollary 3.2, every first eigenfunction is a sum of components odd in exactly one coordinate and even in all the others; each nonzero component is again a first eigenfunction.

A component odd in some y_i would have Rayleigh quotient at least $\lambda_i > \Lambda_n$. In dimension two this follows from Lemma 4.2, since $c_i \geq c_2^\star$; in every dimension $n \geq 3$ it follows from Lemma 4.4, since $c_i \geq 2$. Its quotient is instead $\mu_1(\Omega_c) \leq \Lambda_n$, so every such component is zero. All first eigenfunctions are consequently odd in s and even in $y_1, \ldots, y_m$. It follows that

$$\mu_1(\Omega_c) \geq \lambda_{\|} = \Lambda_n.$$

The reverse inequality was already proved. Finally, Lemma 4.3 identifies the entire first eigenspace with $\operatorname{span}\{U\}$. □

5. Diameter, volume, and independent deformations

We continue to assume the coefficient condition in Theorem 1.1. The diameter of a bounded set is the supremum of the distances between two of its points. Passing to the closure does not change it.

Proposition 5.1 (Exact diameter)**.** *The closure of Ω_c is contained in the Euclidean closed ball of radius $\pi/2$ centered at the origin. Its only points on that sphere are p_+ and p_-. In particular*

$$d(\Omega_c) = \pi. \tag{5.1}$$

Proof. Put $c_{\min} = \min_i c_i$ and $q = \cos s$. For a point of the closure, the defining inequality implies

$$|Y|^2 \leq \frac{q}{c_{\min}} < 1.$$

The function $r \mapsto \arcsin(r)/r$, with its continuous value at zero, is increasing on $[0, 1)$. Indeed, arcsin is convex and vanishes at zero, so its secant slopes from zero are increasing. If $|Y| > 0$, it follows that

$$\sum_i \arcsin^2 |Y_i| \leq \sum_i \frac{|Y_i|^2}{|Y|^2} \arcsin^2 |Y| = \arcsin^2 |Y|.$$

The same inequality is immediate when $Y = 0$. Applying the secant slope inequality once more, now between $\sqrt{q/c_{\min}}$ and $c_{\min}^{-1/2}$, gives

$$|y|^2 \leq 2 \arcsin^2 \sqrt{\frac{q}{c_{\min}}} \leq 2 \arcsin^2(c_{\min}^{-1/2})\, q \leq \frac{\pi^2}{6} \cos s. \tag{5.2}$$

For the last inequality, $c_{\min} \geq \tau_n \geq \tau_2$ and $\arcsin(\tau_2^{-1/2}) = \pi/(2\sqrt{3})$.

Concavity of sine on $[0, \pi/4]$ gives $\sin(|s|/2) \geq \sqrt{2}\,|s|/\pi$ for $|s| \leq \pi/2$. Consequently

$$\cos s = 1 - 2\sin^2(s/2) \leq 1 - \frac{4s^2}{\pi^2}.$$

Together with (5.2), this yields

$$s^2 + |y|^2 \leq \frac{\pi^2}{6} + \frac{s^2}{3} \leq \frac{\pi^2}{4}.$$

If $|s| < \pi/2$, the last inequality is strict. The only closure points with $|s| = \pi/2$ are the two poles, so the assertion about the containing sphere follows. The containing ball bounds the diameter by π, and the two poles have distance π. □

Proposition 5.2 (Volume formula and unbounded diameter–volume ratio)**.** *Let ω_m be the volume of the unit ball $B^m \subset \mathbb{R}^m$, and put*

$$I_m = \int_{-\pi/2}^{\pi/2} (\cos s)^{m/2}\, \mathrm{d}s, \qquad K_m = 2^{m/2}\omega_m I_m. \tag{5.3}$$

Then

$$|\Omega_c| = \frac{2^{m/2}}{\sqrt{\prod_i c_i}} \int_{-\pi/2}^{\pi/2} (\cos s)^{m/2} \int_{B^m} \prod_{i=1}^{m} \left(1 - \frac{\cos s\, z_i^2}{c_i}\right)^{-1/2} \mathrm{d}z\, \mathrm{d}s, \tag{5.4}$$

$$\frac{K_m}{\sqrt{\prod_i c_i}} \le |\Omega_c| \le \frac{K_m}{\sqrt{\prod_i c_i}} \prod_i (1 - c_i^{-1})^{-1/2}. \tag{5.5}$$

In particular, for $c_1 = \cdots = c_m = a \to \infty$,

$$|\Omega_{a\mathbf{1}}| = K_m a^{-m/2}(1 + O(a^{-1})), \qquad \frac{d(\Omega_{a\mathbf{1}})^n}{|\Omega_{a\mathbf{1}}|} \sim \frac{\pi^n}{K_m} a^{m/2} \longrightarrow \infty. \tag{5.6}$$

The implicit constant in $O(a^{-1})$ may depend on m, which is fixed. The ratio also tends to infinity when a single coefficient tends to infinity and all the other coefficients are fixed in the allowed range.

Proof. Use the diffeomorphism in Proposition 2.3 with the rescaled labels $\eta_i = z_i/\sqrt{c_i}$, $z \in B^m$. In these coordinates,

$$Y_i = \sqrt{\frac{\cos s}{c_i}}\, z_i, \qquad y_i = \sqrt{2}\arcsin Y_i.$$

At fixed s, its transverse derivative matrix is diagonal, with ith diagonal entry $\sqrt{2\cos s/c_i}\,(1 - \cos s\, z_i^2/c_i)^{-1/2}$. Since the first coordinate remains s, the full Jacobian is

$$\frac{2^{m/2}(\cos s)^{m/2}}{\sqrt{\prod_i c_i}} \prod_i \left(1 - \frac{\cos s\, z_i^2}{c_i}\right)^{-1/2}.$$

The change-of-variables formula and Fubini's theorem prove (5.4). On the integration region, $0 \le \cos s\, z_i^2/c_i \le 1/c_i < 1$. Bounding each factor between 1 and $(1 - 1/c_i)^{-1/2}$ proves (5.5).

When all coefficients equal a, these bounds become

$$K_m a^{-m/2} \le |\Omega_{a\mathbf{1}}| \le K_m a^{-m/2}(1 - a^{-1})^{-m/2}.$$

Since $(1 - a^{-1})^{-m/2} = 1 + O(a^{-1})$, the volume asymptotic follows. Proposition 5.1 gives the ratio asymptotic. If only $c_i \to \infty$, the two bounds in (5.5) instead show that the volume is bounded above and below by positive constant multiples of $c_i^{-1/2}$ for all sufficiently large c_i. The diameter remains π, proving the final assertion. □

Corollary 5.3 (An explicit three-dimensional family)**.** *For every $a \ge 2$, the domain*

$$\Omega_{a,a} = \left\{(s, y_1, y_2) \in \mathcal{B} : a\left(\sin^2 \frac{y_1}{\sqrt{2}} + \sin^2 \frac{y_2}{\sqrt{2}}\right) < \cos s\right\} \subset \mathbb{R}^3$$

has the simple first eigenvalue $\mu_1 = 2$ and the strict hot-spots property. Its volume and diameter–volume ratio satisfy

$$\frac{4\pi}{a} \le |\Omega_{a,a}| \le \frac{4\pi}{a-1}, \qquad \frac{\pi^2}{4}(a-1) \le \frac{d(\Omega_{a,a})^3}{|\Omega_{a,a}|} \le \frac{\pi^2}{4} a. \tag{5.7}$$

In particular the ratio is asymptotic to $\pi^2 a/4$.

Proof. Here $m = 2$, $\omega_2 = \pi$, and $I_2 = \int_{-\pi/2}^{\pi/2} \cos s \, \mathrm{d}s = 2$, so $K_2 = 4\pi$. Substitute these values into (5.5) and use $d = \pi$. The eigenvalue and hot-spots assertions follow from Propositions 4.5 and 2.2. □

Completion of the proof of Theorem 1.1. Proposition 2.1 proves all the geometric regularity and curvature assertions. The first eigenvalue and its full eigenspace are identified in Proposition 4.5. Together with Proposition 2.2, this proves the critical-point assertions and (1.1) for every nonzero first eigenfunction. Propositions 5.1 and 5.2 prove the remaining diameter and ratio claims. □

5.1. **Independent parameter changes and scaling.** The construction supplies an $(n-1)$-parameter family of exact deformations. If $c(t)$ is any smooth path with every $c_i(t) \geq \tau_n$, the defining function $F_{c(t)}$ depends smoothly on t. Proposition 2.1 gives a nonzero spatial gradient at every boundary point, including the poles. The parameter-dependent implicit function theorem therefore gives local boundary graphs smooth jointly in t and the spatial variables. On each compact parameter interval, the bounds $|Y_i| \leq c_i(t)^{-1/2} \leq \tau_n^{-1/2} < 1$ keep the boundaries away from the transverse faces. The set of pairs (t, x) with t in that interval and $x \in \partial\Omega_{c(t)}$ is closed in a fixed compact box, so it is compact and $|\nabla F_{c(t)}|$ has a positive minimum there. A finite collection of the local graphs therefore covers all these boundaries. Proposition 4.5 then gives the same simple first eigenfunction U and eigenvalue Λ_n at every parameter, without any smallness restriction on the total parameter change.

These shape parameters are independent. The intersection of the boundary with the positive y_i axis occurs at $y_i = b_i(c_i) := \sqrt{2}\arcsin(c_i^{-1/2})$, with all other coordinates zero. It is independent of the other coefficients, and

$$b_i'(c_i) = -\frac{1}{\sqrt{2}\, c_i \sqrt{c_i - 1}} < 0.$$

Thus the map from the coefficient vector to these axis intercepts has an invertible diagonal derivative. The coefficients may be varied independently subject to $c_i \geq \tau_n$; Proposition 5.2 gives the unbounded ratio even when only one coefficient varies.

There is also a free length scale. For $h > 0$ and $x_0 \in \mathbb{R}^n$, the translated and dilated domain $x_0 + h\Omega_c$ has first eigenvalue Λ_n/h^2, eigenfunction $U((x - x_0)/h)$, and diameter $h\pi$. To check the spectral position as well as the eigenvalue equation, for $0 \neq f \in H^1(\Omega_c)$ put $f_h(x) = f((x - x_0)/h)$. This change of variables is a linear bijection between the two H^1 spaces, preserves the zero-mean condition, and gives

$$\frac{\int_{x_0+h\Omega_c} |\nabla f_h|^2}{\int_{x_0+h\Omega_c} |f_h|^2} = h^{-2} \frac{\int_{\Omega_c} |\nabla f|^2}{\int_{\Omega_c} |f|^2}.$$

The variational characterization therefore gives the stated first eigenvalue and identifies its eigenspace with the scaled copy of $\operatorname{span}\{U\}$. The affine map also sends boundary to boundary, multiplies distances by h, and multiplies volume by h^n. It follows that the strict hot-spots property and the ratio $d^n/|\Omega|$ are invariant under this transformation.

6. Classification and rigidity of the Neumann domains

We now classify the domains on which the particular function used in this paper satisfies the Neumann condition. The classification does not require convexity, curvature inequalities, or an a priori description by gradient trajectories. Its hypotheses concern a fixed function in fixed Euclidean coordinates; this distinction is relevant to the spectral conclusions stated below.

Throughout this section, let $n \geq 2$, $m = n - 1$, and set

$$I = (-\pi/2, \pi/2), \qquad \mathcal{B} = I \times (-\pi/\sqrt{2}, \pi/\sqrt{2})^m, \qquad p_\pm = (\pm\pi/2, 0).$$

For $y \in (-\pi/\sqrt{2}, \pi/\sqrt{2})^m$, write

$$Y_i = \sin(y_i/\sqrt{2}), \qquad C_i = \cos(y_i/\sqrt{2}), \qquad P = \prod_{i=1}^{m} C_i, \qquad U(s, y) = \sin s\, P.$$

The function U is defined by the same formula on all of $\mathbb{R}^n$, so its derivatives at faces of the closed principal box are unambiguous.

Theorem 6.1 (Classification by the curvature at one pole)**.** *Let $\Omega \subset \mathcal{B}$ be a nonempty connected domain with C^2 boundary, and suppose that*

$$\partial_\nu U = 0 \quad \text{on } \partial\Omega.$$

Then there is a unique symmetric positive definite $m \times m$ matrix A such that

$$(A^{-1})_{ii} < 1 \qquad (1 \leq i \leq m) \tag{6.1}$$

and

$$\Omega = \Omega_A := \{(s, y) \in \mathcal{B} : F_A(s, y) < 0\}, \qquad F_A(s, y) = Y(y)^T A Y(y) - \cos s. \tag{6.2}$$

Its boundary is real analytic. The only points of its closure on the two axial faces of the closed principal box are p_+ and p_-, and its closure is disjoint from all transverse faces. In the common transverse coordinates at the two poles, its outward second fundamental forms are

$$\mathrm{II}_{p_+} = \mathrm{II}_{p_-} = A. \tag{6.3}$$

In particular, the full second fundamental form at either pole determines the whole domain in these coordinates.

Conversely, every symmetric positive definite A satisfying (6.1) defines by (6.2) a bounded connected domain with real-analytic boundary and $\partial_\nu U = 0$.

Proof. Step 1: exclusion of the transverse faces. Since $\Omega \subset \mathcal{B}$, its closure is compact and is contained in the closed principal box. At a point where its C^1 boundary meets a face of this box, the corresponding coordinate function has a local extremum on the boundary. Its differential therefore vanishes on the tangent plane. The coordinate vector normal to that face must consequently be parallel to the boundary normal. Two distinct faces cannot meet $\partial\Omega$ at the same point, because their normal vectors are linearly independent.

Suppose a boundary point z lies on a transverse face $y_i = \varepsilon\pi/\sqrt{2}$, where $\varepsilon \in \{-1, 1\}$. It lies on no other face, its normal is εe_i, and all $C_j(z)$ for $j \neq i$ are positive. Here and below e_i denotes the unit vector in the y_i direction. The Neumann condition and the identity

$$U_{y_i}(z) = -\frac{\sin s(z)}{\sqrt{2}}\, Y_i(z) \prod_{j \neq i} C_j(z)$$

imply $s(z) = 0$. Thus $\nabla U(z) = 0$. Since e_s is tangent to $\partial\Omega$ at z, choose a C^2 boundary curve γ with $\gamma(0) = z$ and $\gamma'(0) = e_s$. Differentiation of $\nabla U(\gamma(t)) \cdot \nu(\gamma(t)) = 0$ at $t = 0$ gives

$$0 = D^2 U(z)[e_s, \nu(z)],$$

because the term containing $\nabla U(z)$ vanishes. This contradicts

$$U_{sy_i}(z) = -\frac{Y_i(z)}{\sqrt{2}} \prod_{j\neq i} C_j(z) \neq 0.$$

There are consequently no transverse-face contacts. By compactness there is a number $\rho < 1$ such that

$$|Y_i| \leq \rho \quad \text{on } \overline{\Omega}, \qquad 1 \leq i \leq m. \tag{6.4}$$

Step 2: complete trajectories and the equatorial label set. On $\mathcal{B}$, $U_s = \cos s\, P > 0$. The smooth vector field

$$V = \frac{\nabla U}{U_s}$$

has s component equal to one and is tangent to $\partial\Omega \cap \mathcal{B}$. Its local flow preserves this boundary. Indeed, its restriction is a C^1 tangent vector field on a C^2 hypersurface; integral curves constructed in boundary charts agree with the ambient integral curves by uniqueness for ordinary differential equations. An ambient curve starting in Ω cannot meet $\partial\Omega \cap \mathcal{B}$ at a finite time: applying the same uniqueness backward from a contact point would place its preceding segment in the boundary. This assertion concerns contacts inside $\mathcal{B}$, where V is defined.

Such an interior curve extends over every $s \in I$. To see this, suppose its maximal interval within Ω has a finite endpoint $s_* \in I$. On a compact s interval containing s_*, (6.4) confines the curve to a compact subset of $\mathcal{B}$. The vector field and its first derivatives are bounded on a neighborhood of this compact set. The curve has a limit in $\overline{\Omega}$ as $s \to s_*$ and has a unique local continuation. An interior limit extends the curve within Ω; a boundary limit is excluded by the backward-uniqueness argument. Both possibilities contradict the choice of s_*.

The first-integral calculation in Proposition 2.3 depends only on U, so along these curves

$$\frac{\mathrm{d}Y_i}{\mathrm{d}s} = -\frac{1}{2}\tan s\, Y_i, \qquad \eta_i := \frac{Y_i}{\sqrt{\cos s}} \quad \text{is constant.}$$

Define the open equatorial label set

$$D = \{\eta \in (-1,1)^m : (0, \sqrt{2}\arcsin\eta) \in \Omega\},$$

where inverse sine is applied componentwise. It is nonempty because every interior curve crosses $s = 0$. The preceding completeness argument proves

$$\Omega = \Phi(I \times D), \qquad \Phi(s,\eta) = \big(s, \sqrt{2}\arcsin(\sqrt{\cos s}\,\eta)\big). \tag{6.5}$$

The smooth inverse is $(s,y) \mapsto (s, Y/\sqrt{\cos s})$, as in Proposition 2.3; thus Φ is a diffeomorphism from $I \times D$ onto Ω. In particular D is connected, and (6.4) gives $\overline{D} \subset [-\rho,\rho]^m \Subset (-1,1)^m$. Consequently $|Y_i| \leq \rho\sqrt{\cos s}$ throughout Ω, so any sequence in Ω approaching an axial face has $y \to 0$. Conversely, since D is nonempty, a fixed-label curve approaches p_+ and p_- at the respective ends of I. Thus the closure meets the axial faces exactly at the two poles, which belong to $\partial\Omega$.

Step 3: the pole equation is exactly quadratic. Near p_+ put $t = \pi/2 - s$. Since s has its maximum on $\overline{\Omega}$ at p_+, the tangent plane there is horizontal. The C^2 boundary can be written in a small cylinder as

$$t = \varphi(y), \qquad \varphi(0) = 0, \quad D\varphi(0) = 0.$$

Here $\varphi \geq 0$, because every boundary point has $s \leq \pi/2$. The local domain is the side $t > \varphi(y)$. The other possible side would contain points $(t,0)$ with $t < 0$, contrary to $\Omega \subset \mathcal{B}$.

Use local coordinates $q = \cos s = \sin t$ and Y, and define

$$g(Y) = \sin\varphi(\sqrt{2}\arcsin Y).$$

Thus the boundary is $q = g(Y)$ and the local domain is $q > g(Y)$. For $X = \nabla U$, multiplying the trajectory equation above by U_s gives $XY_i = -\frac{1}{2}\sin s\, PY_i$; this identity extends to $q = 0$ by continuity. Together with $Xq = -\sin s\, Pq$, it gives

$$X(q - g(Y)) = -\sin s\, P\left(q - \frac{1}{2}Y \cdot \nabla g(Y)\right).$$

The Neumann condition says that X is tangent to the graph. On that graph $q = g(Y)$, and $\sin s\, P$ is nonzero in this pole neighborhood, including its points with $q = 0$. Hence

$$Y \cdot \nabla g(Y) = 2g(Y) \tag{6.6}$$

for every Y in a sufficiently small ball about zero. For a fixed such Y, the segment $\{rY : 0 \le r \le 1\}$ remains in this ball, and

$$\frac{\mathrm{d}}{\mathrm{d}r}\left(\frac{g(rY)}{r^2}\right) = \frac{rY \cdot \nabla g(rY) - 2g(rY)}{r^3} = 0 \qquad (0 < r \le 1).$$

Thus $g(rY)/r^2$ is constant. Since g is C^2, $g(0) = 0$, and $Dg(0) = 0$, Taylor expansion as $r \downarrow 0$ then yields the exact identity

$$g(Y) = \frac{1}{2}Y^T D^2 g(0) Y = Y^T A Y, \qquad A = D^2\varphi(0). \tag{6.7}$$

The last equality uses $D(\sqrt{2}\arcsin Y)|_{Y=0} = \sqrt{2}\,\mathrm{Id}$ and $D\varphi(0) = 0$, which give $D^2 g(0) = 2D^2\varphi(0)$. In particular A is symmetric, and $g \ge 0$ implies $A \ge 0$.

Step 4: the local graph determines every label. Fix any $\eta \in \mathbb{R}^m$. For all sufficiently small $q > 0$ the point

$$z_q = (\arccos q, \sqrt{2}\arcsin(\sqrt{q}\,\eta))$$

lies in the preceding local cylinder and in $\mathcal{B}$. Equations (6.7) and the local one-sided description give

$$z_q \in \Omega \quad \Longleftrightarrow \quad q > g(\sqrt{q}\,\eta) \quad \Longleftrightarrow \quad \eta^T A \eta < 1.$$

If $z_q \in \Omega$, Step 2 transports it to the equator with the same label. At that equatorial point $Y = \eta$; in particular, even though η was chosen arbitrarily in $\mathbb{R}^m$, the existence of this interior trajectory forces $\eta \in (-1, 1)^m$ and hence $\eta \in D$. Conversely, if $\eta \in D$, (6.5) gives $z_q \in \Omega$. We have therefore proved, for all $\eta \in \mathbb{R}^m$,

$$D = \{\eta : \eta^T A \eta < 1\}. \tag{6.8}$$

Since D is bounded, the positive semidefinite matrix A cannot have a nonzero null vector: the entire line in such a direction would belong to D. Thus A is positive definite. The elementary maximization formula

$$\max_{\eta^T A \eta \le 1} \eta_i^2 = e_i^T A^{-1} e_i = (A^{-1})_{ii}$$

follows from Cauchy–Schwarz in the inner product defined by A, with equality on a multiple of $A^{-1}e_i$. Together with $\overline{D} \subset [-\rho, \rho]^m$, it gives $(A^{-1})_{ii} \le \rho^2 < 1$, proving (6.1). Substitution into (6.5) proves (6.2).

Step 5: converse, analyticity, and the two curvature matrices. Let A now be any symmetric positive definite matrix satisfying (6.1), and put $D_A = \{\eta : \eta^T A \eta < 1\}$. The coordinate maximization formula above shows that $\overline{D}_A \Subset (-1, 1)^m$. Substitution into the leaf map gives

$$F_A(\Phi(s, \eta)) = \cos s\,(\eta^T A \eta - 1),$$

so $\Omega_A = \Phi(I \times D_A)$ is a nonempty connected domain. The map Φ extends continuously to $[-\pi/2, \pi/2] \times \overline{D}_A$. Its compact image is exactly $\overline{\Omega}_A$, since $I \times D_A$ is dense in that closed product. Moreover, $|Y_i|^2 \le (A^{-1})_{ii} < 1$ on this image, and at either axial endpoint $Y = 0$. This proves the asserted separation from the transverse faces and the description of the axial contacts.

Write $D_C = \operatorname{diag}(C_1, \dots, C_m)$. Direct calculation gives

$$\nabla F_A = (\sin s, \sqrt{2} D_C A Y), \qquad \nabla F_A \cdot \nabla U = -\sin s \, P \, F_A.$$

At a boundary point with $s \neq 0$, the first component of ∇F_A is nonzero. At a boundary point with $s = 0$, $Y^T A Y = 1$, so $AY \neq 0$; since every $C_i > 0$, the transverse gradient is then nonzero. Near either pole, on the side outside the axial box one has $\cos s < 0$ and hence $F_A > 0$. Thus the restriction to $\mathcal{B}$ introduces no additional boundary piece, and Ω_A is locally the sublevel set $\{F_A < 0\}$ even at the poles. Therefore F_A is an analytic defining function with nonvanishing gradient everywhere on the boundary. The domain has real-analytic boundary and satisfies the stated Neumann condition.

Near either pole, in the inward axial coordinates $t_+ = \pi/2 - s$ and $t_- = s + \pi/2$, respectively, its graph is

$$t_\pm = \arcsin\big(Y(y)^T A Y(y)\big). \tag{6.9}$$

Since $DY(0) = 2^{-1/2}\mathrm{Id}$, these graphs satisfy $Dt_\pm(0) = 0$ and $D^2 t_\pm(0) = A$. At the poles the outward normals are $\pm e_s$, so the second fundamental forms are precisely the Hessians of these inward pole graphs, by (2.4). This proves (6.3). It also proves uniqueness of A. This completes the proof of the classification and, in particular, of Theorem 1.2. □

Corollary 6.2 (Symmetries and the diagonal classification)**.** *Every domain in Theorem 6.1 is invariant under $s \mapsto -s$ and under $y \mapsto -y$. It is invariant under each separate transverse coordinate reflection if and only if A is diagonal. In that case it is exactly the diagonal domain Ω_c with*

$$A = \operatorname{diag}(c_1, \dots, c_m), \qquad c_i > 1.$$

The larger geometric classification $c_i > 1$ does not replace the spectral hypothesis $c_i \geq \tau_n$ of Theorem 1.1. Under that latter hypothesis the first positive Neumann eigenvalue is simple, equals $(n+1)/2$, and has the strict hot-spots property established there.

Proof. The two asserted automatic symmetries follow immediately from (6.2). Let R_i be reflection in the ith transverse coordinate. Since $Y(R_i y) = R_i Y(y)$, invariance is equivalent to invariance of the label ellipsoid under R_i, and hence to $R_i^T A R_i = A$. Equality of the ellipsoids implies equality of their quadratic forms, for example by comparing their radial boundary points in each direction. The matrix identity forces $A_{ij} = 0$ for every $j \neq i$. Requiring it for all i is therefore exactly diagonality. For a positive diagonal matrix, (6.1) is equivalent to $c_i > 1$. The final spectral assertion is precisely Theorem 1.1. □

Corollary 6.3 (Determination by curvature data)**.** *Two domains satisfying Theorem 6.1 in the same principal coordinates are equal if their full second fundamental forms at the upper pole agree. The same conclusion holds using the lower pole.*

In the diagonal range $c_i \geq \tau_n$, two such domains are Euclidean congruent if and only if their unordered lists of pole principal curvatures agree. Equivalently, their coefficient lists agree up to permutation.

Proof. The first assertion follows from (6.3) and the uniqueness of A. For the second, equal coefficient lists up to permutation give congruent domains by the corresponding transverse coordinate permutation. Conversely, Proposition 5.1 shows that the only points of the closure on the sphere of radius $\pi/2$ are the poles. If two closure points have distance π, equality in the triangle inequality forces both of them onto this sphere. Thus the pole pair is the unique pair realizing the diameter. A congruence must preserve this unordered pair, and hence it preserves the unordered list of principal curvatures at a pole. By (6.3) this list is $\{c_1, \dots, c_m\}$ at either pole, proving necessity. □

Proposition 6.4 (All exact deformations preserving the fixed function)**.** *Let $J \subset \mathbb{R}$ be an open interval. A smooth family of domains $\Omega_t \subset \mathcal{B}$ satisfying the hypotheses of Theorem 6.1 for the same function U is exactly a family $\Omega_{A(t)}$, where $A(t)$ is a smooth path of symmetric positive definite matrices satisfying (6.1). Here a smooth family means that its local boundary graphs depend smoothly on the parameter and on the spatial variables.*

Its outward normal velocity at a point $(s, y) \in \partial\Omega_{A(t)}$ is

$$v_\nu(t, s, y) = -\frac{Y(y)^T \dot{A}(t) Y(y)}{|\nabla F_{A(t)}(s, y)|}. \tag{6.10}$$

If every separate coordinate reflection is preserved, the paths are exactly the diagonal paths $A(t) = \operatorname{diag}(c_1(t), \dots, c_m(t))$ with $c_i(t) > 1$. If the full second fundamental form at either pole is independent of t, then the entire family of domains is constant. In particular, the deformations preserving the fixed Neumann function have $m(m+1)/2$ matrix parameters, or m parameters when all these reflections are required.

Proof. Apply Theorem 6.1 at each parameter. The matrix is the Hessian of the upper inward pole graph, so smooth dependence of the local boundary graphs implies smooth dependence of $A(t)$. Conversely, $F_{A(t)}$ depends smoothly on (t, s, y), and the proof of the theorem gives a nonvanishing spatial gradient on every boundary. The implicit function theorem therefore gives smooth local boundary graphs. On each compact parameter interval, continuity of $A(t)$ and $A(t)^{-1}$ gives constants $a_0 > 0$ and $\rho < 1$ such that $A(t) \succeq a_0 \mathrm{Id}$ and $(A(t)^{-1})_{ii} \le \rho^2$. Thus the compactness argument used for parameter changes in Section 5 applies here as well: the boundaries remain separated from the transverse faces and are covered by finitely many smooth local graphs over that interval. This proves the converse smooth-family assertion.

For any local parametrization $z(t)$ of a moving boundary point, differentiation of $F_{A(t)}(z(t)) = 0$ gives

$$0 = Y^T \dot{A}(t) Y + \nabla F_{A(t)} \cdot \dot{z}(t).$$

Since $\nu = \nabla F_{A(t)}/|\nabla F_{A(t)}|$, this is (6.10), independently of tangential choices in the parametrization. The reflection assertion follows from Corollary 6.2. The assertion about fixed pole curvature follows from $A(t) = \mathrm{II}_{p_+}(t) = \mathrm{II}_{p_-}(t)$ and the uniqueness in the classification theorem. Finally, the admissible matrices form an open subset of the vector space of symmetric matrices, and the admissible diagonal matrices form an open subset of the diagonal subspace. Their dimensions are the two stated numbers. □

Remark 6.5 (The spectral scope of the classification). The ambient equation established in Proposition 2.2 gives $-\Delta U = ((n+1)/2)U$ on every classified domain. The classification supplies the Neumann boundary condition and the geometry of all its possible domains in the principal box. It does not assert that this eigenvalue is first for every admissible matrix, or for every diagonal list $c_i > 1$. For diagonal paths satisfying $c_i(t) \ge \tau_n$, Theorem 1.1 supplies that additional conclusion at every parameter: U remains the simple first eigenfunction, and its two hot spots remain the poles. Convexity and curvature assumptions are likewise absent from the classification theorem; the geometric conclusions required for the main hot-spots theorem have already been proved in its stated coefficient range.

Remark 6.6 (The first eigenvalue and diameter do not determine the domain). The curvature determination above cannot be replaced by a determination from the first eigenvalue and diameter. Set $A = a\,\mathrm{Id}$ with $a \ge 2$. All these diagonal domains belong to the range of Theorem 1.1, and all have

$$\mu_1(\Omega_{a\,\mathrm{Id}}) = \frac{n+1}{2}, \qquad d(\Omega_{a\,\mathrm{Id}}) = \pi.$$

For $b > a$, however, $\Omega_{b\,\mathrm{Id}} \subsetneq \Omega_{a\,\mathrm{Id}}$, and the difference contains a nonempty open set: one can choose $s = 0$, $Y_2 = \cdots = Y_m = 0$, and $b^{-1} < Y_1^2 < a^{-1}$, then take a sufficiently small neighborhood. Their volumes are therefore strictly different, so the domains are not congruent. Thus even the simultaneous values of μ_1, d, and $\mu_1 d^2$, together with the fixed locations of the two hot spots, leave nontrivial deformations.

7. Pole regularity for general leaf-label sets

7.1. **A local classification without lateral smoothness assumptions.** The classification in Section 6 begins with a globally C^2 Neumann domain and derives the foliation. If the foliation is prescribed from the start, smoothness at one pole alone forces the label set to be an ellipsoid. The following local statement imposes no regularity on the lateral boundary and no convexity or connectedness assumption on the label set.

Proposition 7.1 (A twice differentiable pole forces quadratic labels)**.** *Let $D \subset \mathbb{R}^m$ be a bounded open set containing the origin, with $\overline{D} \subset (-1,1)^m$. Define*

$$\Omega(D) = \left\{ \left(s, \left(\sqrt{2}\arcsin(\eta_i\sqrt{\cos s})\right)_{i=1}^m \right) : |s| < \pi/2,\ \eta \in D \right\}. \tag{7.1}$$

Suppose that $\partial\Omega(D)$ is a C^2 hypersurface near $p_+ = (\pi/2, 0)$. Then there is a positive definite symmetric matrix H such that

$$D = \{\eta \in \mathbb{R}^m : \eta^T H \eta < 1\}. \tag{7.2}$$

More precisely, H is the Hessian at zero of the inward pole graph $\pi/2 - s = \varphi(y)$.

Proof. The point p_+ is in the boundary: the label $\eta = 0$ gives interior points approaching it, and the whole set lies in $s < \pi/2$. On a C^2 boundary near that point, the restriction of s has a maximum at p_+, so the tangent plane is $s = \pi/2$. The graph representation theorem therefore gives

$$t := \pi/2 - s = \varphi(y), \qquad \varphi(0) = 0, \qquad D\varphi(0) = 0,$$

where φ is C^2. Let $H = D^2\varphi(0)$. Once this graph exists, only its second-order Taylor expansion at zero will be used.

Fix $\eta \in \partial D$. For every sufficiently small $t > 0$, the point with $s = \pi/2 - t$ and

$$y_i(t) = \sqrt{2}\arcsin(\eta_i\sqrt{\sin t})$$

belongs to $\partial\Omega(D)$. Indeed, the leaf-coordinate map from Proposition 2.3 is a diffeomorphism on a neighborhood of $I \times \overline{D}$, with the inverse formula given there. It therefore takes $I \times \partial D$ to the nonpolar boundary of $\Omega(D)$. The point converges to p_+ and eventually lies in the pole graph neighborhood.

For this fixed label, Taylor expansion yields

$$y(t) = \sqrt{2t}\,\eta + O(t^{3/2}).$$

Since $t = \varphi(y(t))$, the second-order Taylor formula for φ gives

$$t = \frac{1}{2} y(t)^T H y(t) + o(|y(t)|^2) = t\,\eta^T H \eta + o(t).$$

Divide by t and let $t \downarrow 0$. We obtain

$$\eta^T H \eta = 1 \qquad \text{for every } \eta \in \partial D. \tag{7.3}$$

We now show that this identity alone forces (7.2), including positive definiteness. For any unit vector $\theta \in \mathbb{R}^m$, let

$$r_\theta = \sup\{r > 0 : \rho\theta \in D \text{ for all } 0 \le \rho < r\}.$$

Because D contains a neighborhood of zero and is bounded, $0 < r_\theta < \infty$. The segment $\{r\theta : 0 \le r < r_\theta\}$ lies in D, so its endpoint is in $\overline{D}$. If that endpoint belonged to D, openness would extend the segment beyond r_θ, contradicting its definition. Hence $r_\theta\theta \in \partial D$. Equation (7.3) implies

$$\theta^T H\theta = r_\theta^{-2} > 0.$$

Thus H is positive definite. The unique positive intersection of this ray with the quadric $\eta^T H\eta = 1$ is precisely $r_\theta\theta$. The definition of r_θ already shows that the ellipsoid $E = \{\eta : \eta^T H\eta < 1\}$ is contained in D.

If a point $\rho\theta \in D$ lay outside $\overline{E}$, then $\rho > r_\theta$. Since D is open and bounded, the component of $\{r > 0 : r\theta \in D\}$ containing ρ would have a finite right endpoint $R > \rho$, with $R\theta \in \partial D$ by the same endpoint argument. Equation (7.3) would imply $R = r_\theta$, a contradiction. Hence $D \subset \overline{E}$. Openness excludes points on ∂E as well, since every neighborhood of such a point meets the complement of $\overline{E}$. Therefore $D = E$, as claimed. □

For the diagonal label set D_c, the matrix is $H = \operatorname{diag}(c_i)$, in agreement with (2.6) and Theorem 6.1. The next example shows why twice differentiable pole regularity is needed even when the label set is smooth, convex, and invariant under every coordinate reflection.

7.2. **Sharpness of twice differentiable pole regularity.** The regularity assumption at a pole has a precise role in the classification: it forces the homogeneous label function to be a quadratic form. The following example shows that $C^{1,1}$ regularity does not suffice, even for convex domains with all coordinate reflection symmetries.

Proposition 7.2 (A convex example with nonquadratic labels)**.** *Suppose $m = n - 1 \ge 2$. For $\eta \in \mathbb{R}^m$ set*

$$h(\eta) = \left(\sum_{i=1}^m \eta_i^4\right)^{1/2}, \qquad Q(\eta) = 3|\eta|^2 + h(\eta), \qquad D_Q = \{\eta \in \mathbb{R}^m : Q(\eta) < 1\}. \tag{7.4}$$

Using the transverse coordinates $Y_i = \sin(y_i/\sqrt{2})$, define

$$F_Q(s, y) = Q(Y) - \cos s, \qquad \Omega_Q = \{(s, y) \in \mathcal{B} : F_Q(s, y) < 0\}. \tag{7.5}$$

Then Ω_Q is a bounded convex domain with $C^{1,1}$ boundary, invariant under reflection in every coordinate hyperplane. Its boundary is real analytic and has strictly positive principal curvatures away from $p_\pm = (\pm\pi/2, 0)$. At either pole the boundary is not twice differentiable.

The function $U(s, y) = \sin s \prod_i \cos(y_i/\sqrt{2})$ satisfies

$$-\Delta U = \Lambda_n U \quad \text{in } \Omega_Q, \qquad \partial_\nu U = 0 \quad \text{on } \partial\Omega_Q, \qquad \int_{\Omega_Q} U = 0. \tag{7.6}$$

The domain Ω_Q is obtained by sweeping the labels in D_Q along the gradient leaves of U, but D_Q is not an ellipsoid.

Proof. We first verify the regularity and convexity of the label function. The function h is homogeneous of degree two, is real analytic away from the origin, and satisfies

$$0 \le h(\eta) \le |\eta|^2.$$

In particular it is differentiable at the origin with gradient zero. For $\eta \ne 0$, put $a = (\eta_1^3, \dots, \eta_m^3)^T$. Direct differentiation gives

$$h_i = \frac{2\eta_i^3}{h}, \qquad D^2 h = \frac{6}{h}\operatorname{diag}(\eta_i^2) - \frac{4}{h^3} aa^T. \tag{7.7}$$

For every $\xi \in \mathbb{R}^m$, Cauchy–Schwarz gives

$$\left(\sum_i \eta_i^3 \xi_i\right)^2 \le \left(\sum_i \eta_i^4\right)\left(\sum_i \eta_i^2 \xi_i^2\right).$$

Consequently

$$0 \le \frac{2}{h}\sum_i \eta_i^2\xi_i^2 \le D^2h[\xi,\xi] \le \frac{6}{h}\sum_i \eta_i^2\xi_i^2 \le 6|\xi|^2, \tag{7.8}$$

where the last step uses $\eta_i^2 \le h$. The derivative formula and $\sum_i \eta_i^6 \le |\eta|^2 \sum_i \eta_i^4$ give $|\nabla h(\eta)| \le 2|\eta|$, so the gradient is continuous at the origin. Integrating the Hessian bound along a line segment that avoids the origin shows that the gradient is 6-Lipschitz. If a segment passes through the origin, apply the same estimate on its two parts and pass to the origin by continuity; the lengths of the parts add to the length of the segment. Thus $h \in C^{1,1}$ on $\mathbb{R}^m$. Along every line, the first derivative of h is continuous and has nonnegative derivative away from a possible passage through the origin, by (7.8). It is therefore nondecreasing on the whole line, proving convexity. It follows that Q is $C^{1,1}$, is convex, and has $D^2Q \succeq 6I$ away from the origin.

We next prove convexity in the original Euclidean coordinates. Let

$$b = \sqrt{2}\arcsin(1/\sqrt{3}), \qquad K = [-\pi/2, \pi/2] \times [-b, b]^m.$$

The inequality $Q(Y) \ge 3|Y|^2$ implies

$$\Omega_Q = \operatorname{int} K \cap \{F_Q < 0\}.$$

On K one has $Y_i^2 \le 1/3$. Write $C = \operatorname{diag}(C_1, \dots, C_m)$, where $C_i = \cos(y_i/\sqrt{2})$. At points with $Y \ne 0$, the chain rule and (7.7) give

$$D_y^2(h(Y(y))) = \frac{1}{2}CD_Y^2h(Y)C - \operatorname{diag}\left(\frac{Y_i^4}{h(Y)}\right) \succeq -\frac{1}{3}I.$$

Indeed, the first term is nonnegative, and $Y_i^4/h(Y) \le Y_i^2 \le 1/3$. The quadratic part in the label coordinates satisfies

$$D_y^2\left(3\sum_i Y_i^2\right) = \operatorname{diag}(3 - 6Y_i^2) \succeq I.$$

These estimates hold at every point of K with $Y \ne 0$, and hence almost everywhere on K:

$$D_y^2 F_Q \succeq \frac{2}{3}I, \qquad (F_Q)_{ss} = \cos s \ge 0, \qquad (F_Q)_{sy_i} = 0. \tag{7.9}$$

The function F_Q is $C^{1,1}$ on a neighborhood of K. For such a function, a nonnegative Hessian almost everywhere on a convex open set implies convexity there: its distributional Hessian is the almost everywhere Hessian, convolution on compact subsets gives smooth functions with nonnegative Hessian, and passage to the limit gives the convexity inequality on every segment. Continuity includes the boundary of K. Thus F_Q is convex on K, and Ω_Q is convex. It is nonempty because $F_Q(0,0) = -1$, and it is bounded because it lies in K. In particular it is a connected domain. The evenness of Q in each coordinate and of $\cos s$ gives all the stated reflections.

For completeness, its closure is exactly $K \cap \{F_Q \le 0\}$: each point of this set is approached from $F_Q < 0$ by taking its convex combinations with $(0,0)$, where $F_Q = -1$. There is no contact with a transverse face of K, because $h(Y) \ge Y_i^2$ implies $Q(Y) \ge 4Y_i^2$, whereas such a face has $Y_i^2 = 1/3$. On either axial face, $F_Q \le 0$ forces $Y = 0$. Thus the only axial-face contacts are p_+ and p_-.

We check boundary regularity, including these contacts. On $F_Q = 0$ with $s \ne 0$, including the poles, the derivative $(F_Q)_s = \sin s$ is nonzero. At a point with $s = 0$, one has $Q(Y) = 1$; Euler's identity

$$Y \cdot \nabla Q(Y) = 2Q(Y) \tag{7.10}$$

then shows that $\nabla Q(Y) \neq 0$. Since every $C_i > 0$, the identity

$$\nabla_y F_Q = \frac{1}{\sqrt{2}} C \nabla Q(Y)$$

shows that $\nabla F_Q \neq 0$ there as well. The C^1 implicit function theorem gives a C^1 graph at every boundary point. To obtain $C^{1,1}$ regularity, differentiate the graph identity: its derivatives are quotients of first derivatives of F_Q evaluated on the graph, with a denominator bounded away from zero. The C^1 graph is locally Lipschitz, and the first derivatives of F_Q are locally Lipschitz, so these quotients are locally Lipschitz as well. Near the upper pole, points with $s > \pi/2$ have $F_Q > 0$; the corresponding statement holds below the lower pole. Therefore the principal-box restriction contributes no additional boundary near either pole.

At every nonpolar boundary point, $Y \neq 0$ and $\cos s = Q(Y) > 0$. Hence F_Q is real analytic there and its Hessian is positive definite by (7.9). For a tangent vector $T \neq 0$, the outward second fundamental form is $D^2 F_Q[T,T]/|\nabla F_Q| > 0$. This proves the asserted regularity and positive principal curvatures away from the poles.

We now verify the Neumann condition and the label description. The ambient equation $-\Delta U = \Lambda_n U$ was established in Proposition 2.2. Using the gradient of U computed there, the chain rule and (7.10) yield the exact identity

$$\begin{aligned} \nabla F_Q \cdot \nabla U &= \sin s \, P \left(\cos s - \frac{1}{2} Y \cdot \nabla Q(Y) \right) \\ &= -U F_Q. \end{aligned} \tag{7.11}$$

All terms are continuous even at $Y = 0$, where $\nabla Q(0) = 0$. Since the normal is $\nu = \nabla F_Q / |\nabla F_Q|$ on the entire $C^{1,1}$ boundary, (7.11) proves $\partial_\nu U = 0$, including the poles. The zero mean follows from oddness in s.

For $|s| < \pi/2$, introduce the leaf labels $\eta_i = Y_i/\sqrt{\cos s}$. Homogeneity gives $Q(Y) = \cos s \, Q(\eta)$, so $F_Q < 0$ is equivalent to $\eta \in D_Q$. Also $Q(\eta) \geq 4\eta_i^2$ gives $\overline{D}_Q \subset [-1/2, 1/2]^m$. The label set is convex because Q is convex, and its boundary is real analytic: Q is analytic away from zero, while (7.10) gives $\nabla Q \neq 0$ on $Q = 1$. Consequently $\Omega_Q = \Omega(D_Q)$ in (7.1). Proposition 2.3 identifies its fixed-label curves as gradient trajectories of U; that trajectory calculation depends only on the ambient function.

Finally, near the upper pole write $t = \pi/2 - s$. The boundary is the graph

$$t = \varphi(y) = \arcsin Q(\sin(y/\sqrt{2})) = \frac{1}{2} \left(3|y|^2 + h(y) \right) + O(|y|^4). \tag{7.12}$$

To justify the remainder, note that $\sin(y/\sqrt{2}) = y/\sqrt{2} + O(|y|^3)$ componentwise and $|\nabla Q(\eta)| \leq 8|\eta|$. The change in Q is therefore $O(|y|^4)$; its homogeneity gives $Q(y/\sqrt{2}) = Q(y)/2$, and the remainder in $\arcsin q = q + O(q^3)$ contributes only $O(|y|^6)$. If φ were twice differentiable at zero, comparison of its second-order Taylor expansion with (7.12) at $y = rv$, after division by r^2 and passage to $r \downarrow 0$, would give $Q(v) = v^T H v$ for every $v \in \mathbb{R}^m$, where $H = D^2\varphi(0)$. That is impossible when $m \geq 2$. Indeed,

$$Q(e_1) = Q(e_2) = 4, \qquad Q(e_1 + e_2) = Q(e_1 - e_2) = 6 + \sqrt{2},$$

which violates the parallelogram identity required of a quadratic form:

$$Q(e_1 + e_2) + Q(e_1 - e_2) = 12 + 2\sqrt{2} \neq 16 = 2Q(e_1) + 2Q(e_2).$$

Reflection in $s = 0$ gives the same failure at the lower pole. If D_Q were an ellipsoid, its central symmetry would force its center to be zero. Write it as $\{\eta : \eta^T B \eta < 1\}$ with B positive definite. On the ray rv, $r > 0$, for any $v \neq 0$, positive degree-two homogeneity places the boundary at $r = Q(v)^{-1/2}$, while the ellipsoid places it at $r = (v^T B v)^{-1/2}$. Equality of these parameters would

give $Q(v) = v^T B v$, contradicting the same parallelogram calculation. Thus the label set is not an ellipsoid. □

This proposition establishes sharpness of the pole regularity in the fixed-U classification of Theorem 6.1 and Proposition 7.1. Its conclusion concerns the geometry and the Neumann condition; the spectral position of the eigenfunction in (7.6) is not asserted.

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